integer, as follows:

```
GroupKeyHash =
              Crypto_KDF
              (
                  InputKey = OperationalGroupKey,
                  Salt = [],
                  Info = "GroupKeyHash",
                  Length = 16  // Bits
             )

// GroupKeyHash is an array of 2 bytes (16 bits) per Crypto_KDF

// GroupSessionId is computed by considering the GroupKeyHash as a Big-Endian
// value. GroupSessionId is a scalar. Its use in fields within messages may cause a
// re-serialization into a different byte order than the one used for initial
generation.
GroupSessionId = (GroupKeyHash[0] << 8) | (GroupKeyHash[1] << 0)
```

where `[]` denotes a zero-length array.

For example, given:

- An Operational Group Key value of: `a6:f5:30:6b:af:6d:05:0a:f2:3b:a4:bd:6b:9d:d9:60`

After the above derivation following the definition of `Crypto_KDF` in Section 3.8, “Key Derivation Function (KDF)”, the resulting Group Session ID data would be:

- Raw output of GroupKeyHash: `b9:f7`
- Group Session ID scalar value to be used for further processing: 0xB9F7 (47607 decimal)

The *Group Session ID* MAY help receiving nodes efficiently locate the *Operational Group Key* used to encrypt an incoming groupcast message. It SHALL NOT be used as the sole means to locate the associated *Operational Group Key*, since it MAY collide within the fabric. Instead, the *Group Session ID* provides receiving nodes a means to identify *Operational Group Key* candidates without the need to first attempt to decrypt groupcast messages using all available keys.

On receipt of a message of Group Session Type, all valid, installed, operational group key candidates referenced by the given Group Session ID SHALL be attempted until authentication is passed or there are no more operational group keys to try. This is done because the same Group Session ID might arise from different keys. The chance of a Group Session ID collision is 2^{-16} but the chance of both a Group Session ID collision and the message MIC matching two different operational group keys is 2^{-80}.

Group Session Ids are sized to fit within the context of the Session Identifier field of a message. When used in this context, the Group Session ID value allows a receiving Node to identify the appropriate message encryption key to use from the set of active operational keys it has currently

 Copyright © Connectivity Standards Alliance, Inc. All rights reserved.